SOC FINAL INCIDENT REPORT

Lab 5: Comprehensive Threat Investigation

1. Incident Overview & Triage Metadata

Incident Title	Suspicious Office-to-PowerShell Execution, LSASS Access, Registry Persistence & Network Staging
Assigned Severity	**CRITICAL — simulated lab scenario assessment**
Affected Host	HR-WKSTN-05 (some indexed records display host value **svg**)
Associated Account	HR-WKSTN-05\s.patel (as represented in the lab dataset)
Observed Window	2026-10-14; approximately 09:55–10:06 (+05:30)
Detection Source	Splunk Enterprise — index="soc-investigation"; Lab 5 synthetic investigation dataset
Evidence Scope	Synthetic endpoint telemetry and synthetic PCAP; documentation-range IP addresses and example.test domain

2. Executive Summary

Lab 5 contains a simulated multi-stage endpoint investigation involving suspicious Office-to-PowerShell execution, LSASS process access, a registry Run-key persistence-like event, archive staging, and outbound network activity. Process telemetry includes **WINWORD.EXE** launching **powershell.exe** with hidden, execution-policy-bypass, and encoded-command characteristics. Additional records show **rundll32.exe** with **comsvcs.dll MiniDump**-style arguments targeting **lsass.exe**, followed by archive preparation and **curl.exe** network activity to port 8443.

A registry telemetry record at approximately 10:04 identifies **update_service.exe** and describes a persistence-like Run-key value being set. The synthetic PCAP shows a GET for **/stage/update_service.exe** returning placeholder content **SYNTHETIC-DEMO!!**, and a POST to **/upload** containing the lab marker **diagnostic_bundle.cab** and **SYNTHETIC_ONLY**.

Evidence limitations: PCAP timestamps do not align closely enough with Splunk process telemetry to establish exact process-to-network attribution. The POST body names an archive but does not contain the archive itself; HTTP 202 does not prove successful exfiltration. Addresses 198.51.100.0/24, 203.0.113.0/24 and 192.0.2.0/24 are reserved documentation ranges, and .example.test is synthetic. This is a simulated scenario, not a verified compromise or real malicious infrastructure.

3. Reconstructed Process Activity & Investigation Timeline

Observed Activity	Evidence / Interpretation
Office → PowerShell	WINWORD.EXE is the parent of powershell.exe. Command-line indicators include hidden execution, bypass, and encoded-command behavior. Suspicious execution pattern.
LSASS access	rundll32.exe with comsvcs.dll MiniDump-style arguments targets lsass.exe. Consistent with an attempted LSASS dump; successful dump creation is not independently established.
Archive staging	makecab.exe appears in simulated process activity; diagnostic_bundle.cab is the referenced archive artifact. The PCAP POST does not show archive contents.
Outbound activity	Splunk records include curl.exe communicating with 198.51.100.99:8443. Exact correlation to the PCAP POST is not established because timestamps differ.
Registry event	At approximately 10:04, EventCode 13 records update_service.exe and the description “Persistence-like Run key value set; simulated registry telemetry.” Exact registry path/value is unavailable.
PCAP HTTP GET	GET /stage/update_service.exe to 198.51.100.99; response 200 OK with placeholder body SYNTHETIC-DEMO!!.
PCAP HTTP POST	POST /upload to 198.51.100.99:8443; form fields include lab=05, artifact=diagnostic_bundle.cab, note=SYNTHETIC_ONLY; response 202 Accepted. Not proof of archive transfer.

4. MITRE ATT&CK Mapping (Candidate)

Tactic	Technique ID & Name	Supporting Lab Evidence	Confidence
Execution	**T1059.001** — PowerShell	PowerShell launched by WINWORD.EXE with hidden, bypass, and encoded-command indicators.	High for observed behavior
Credential Access	**T1003.001** — OS Credential Dumping: LSASS Memory	rundll32.exe / comsvcs.dll MiniDump-style activity targeting lsass.exe.	Attempt indicated; outcome unverified
Persistence	**T1547.001** — Registry Run Keys / Startup Folder	EventCode 13 describes a simulated Run-key value set involving update_service.exe; exact path/value absent.	Candidate; simulated record
Collection	**T1560.001** — Archive Collected Data: Archive via Utility	makecab.exe activity and diagnostic_bundle.cab artifact reference.	Candidate
Command and Control	**T1071.001** — Web Protocols	Synthetic HTTP GET/POST in PCAP; endpoint attribution not conclusively correlated.	Candidate; synthetic PCAP
Exfiltration	**T1041** — Exfiltration Over C2 Channel	POST references diagnostic_bundle.cab; archive bytes are not present in displayed form body.	Unconfirmed; do not claim success

5. Indicators & Evidence References

Indicator Type	Observed Value	Context / Handling
Documentation IPv4	198.51.100.99	Synthetic endpoint in lab network records and PCAP; not a real threat IOC.
Documentation IPv4	203.0.113.25	Synthetic destination in network telemetry; not a real threat IOC.
Documentation IPv4	192.0.2.55	Synthetic PCAP source address.
Documentation IPv4:Port	198.51.100.99:8443	Synthetic HTTP POST destination; 202 response does not prove archive transfer.
Domain	fileserver.example.test	Synthetic example domain; not a live domain indicator.
Processes / Artifact	powershell.exe; rundll32.exe; update_service.exe; makecab.exe; curl.exe; diagnostic_bundle.cab	Simulated process and artifact names in Lab 5 dataset.
HTTP markers	/stage/update_service.exe; /upload; SYNTHETIC-DEMO!!; SYNTHETIC_ONLY	Synthetic PCAP paths and placeholder markers.

6. Analyst Assessment & Recommended Response

Assessment: The dataset simulates high-risk behaviors investigated by a SOC: Office-spawned encoded PowerShell, LSASS access consistent with a dump attempt, registry persistence-like activity, archive staging, and HTTP network activity. The scenario may be rated **Critical for lab demonstration purposes**, but the available evidence does not establish a real endpoint compromise, successful credential extraction, or successful data exfiltration.

Final conclusion: Lab 5 demonstrates a multi-stage synthetic threat-investigation scenario and supports candidate MITRE ATT&CK mappings for execution, credential access, persistence, collection, and web-protocol communication. Exfiltration remains unconfirmed. Reserved IP addresses, example domains, placeholder payloads, and simulated telemetry must remain labeled as training evidence.